

Agenda.

- Type of LLD interviews.
- How to approach LLD interviews.
- Design a Pen.

Type of low level Design interviews.

① Theoretical

- ⇒ Questions around OOPS, Design Patterns & Principles language concepts etc.
- ⇒ Knowledge sound on LLD.
- ⇒ Older tech companies
(Big MNC's, banks)
- ⇒ ~ 45 mins

② Design Rounds.

- ⇒ Problem statement to design some system.
- ⇒ Case study based discussion.
- ⇒ Big MNC's: Amazon / Walmart / Microsoft / Paypal... -
- ⇒ ~ 45 mins
- ⇒ Some code might be expected.

③ Machine Coding.

⇒ ~2.5 - 3 hrs.

⇒ Case Study.

⇒ Design + Working Code.

⇒ write end to end working code for the given set of requirements.

⇒ Almost every top tech startups even for freshers.
Flipkart / CRED / Swiggy / PhonePe / Zepto /

Design & Machine Coding rounds.

(~45-50 mins)
Design

→ Single line problem statement.

→ Case Study

1) Gather requirements. (5-10 mins)

2) Class Program (5-10 mins)

3) Schema Design (5-10 mins)

4) Write basic code structure for some requirement.
(10-15 mins)

5) follow ups

Machine Coding

⇒ Design + Working Code
round

1) Gather requirements.

2) Class Program

3) Schema Design

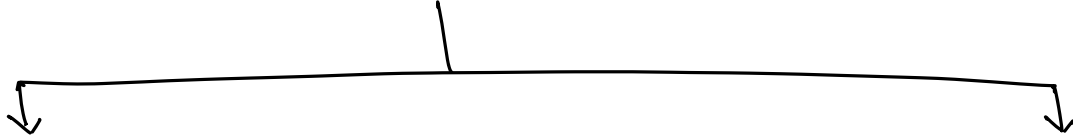
4) Write end to end working code. (1-1.5 hours)

5) follow ups

30 mins

How to Approach a Design Problem.

① Get an overview of the system.



We already know about the system

We don't know about the system.

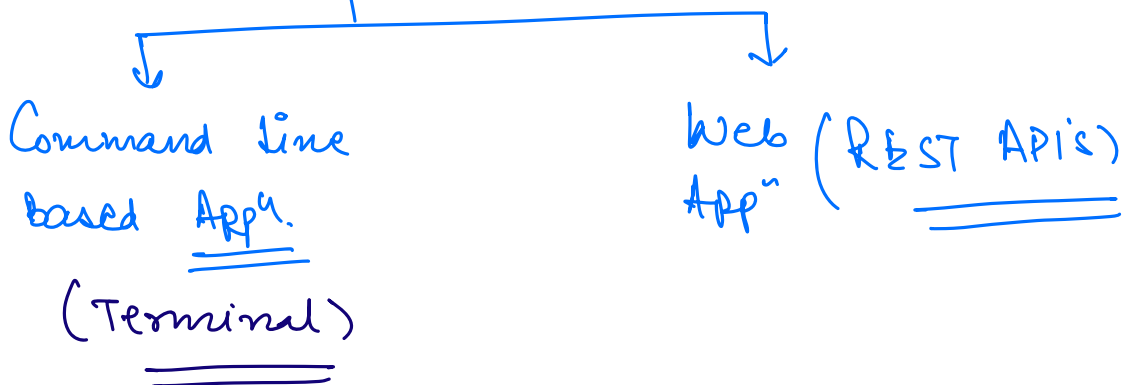
⇒ Align yourself with the interviewers expectations.

⇒ Ask the interviewer to explain about the system briefly.

What kind of system we need to design?

→ Entity (Requirements + Class Diagram)

→ Working App



Do we need to store data in Database?

↳ Yes or No.

② Gathering requirements.

⇒ 5-8 core features.

⇒ Try to visualize the system & suggest some of the important features.

⇒ { Try to find out the edges cases | corner cases.

→ Clarify requirements.

(figuring out the Edge | corner cases)

③ Class Diagram.

→ Classes / Interfaces
→ Relⁿ b/w them
→ Design Patterns.

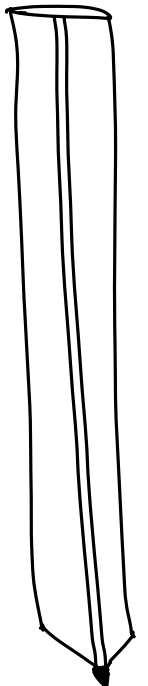
④ Schema Design.

Design a Pen.

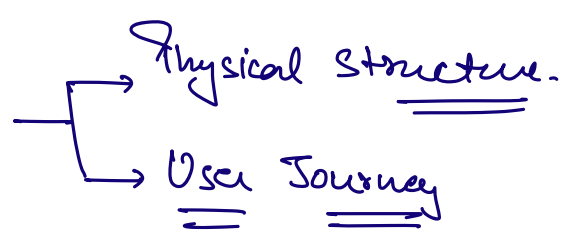
⇒ Design PEN entity.

Requirements.

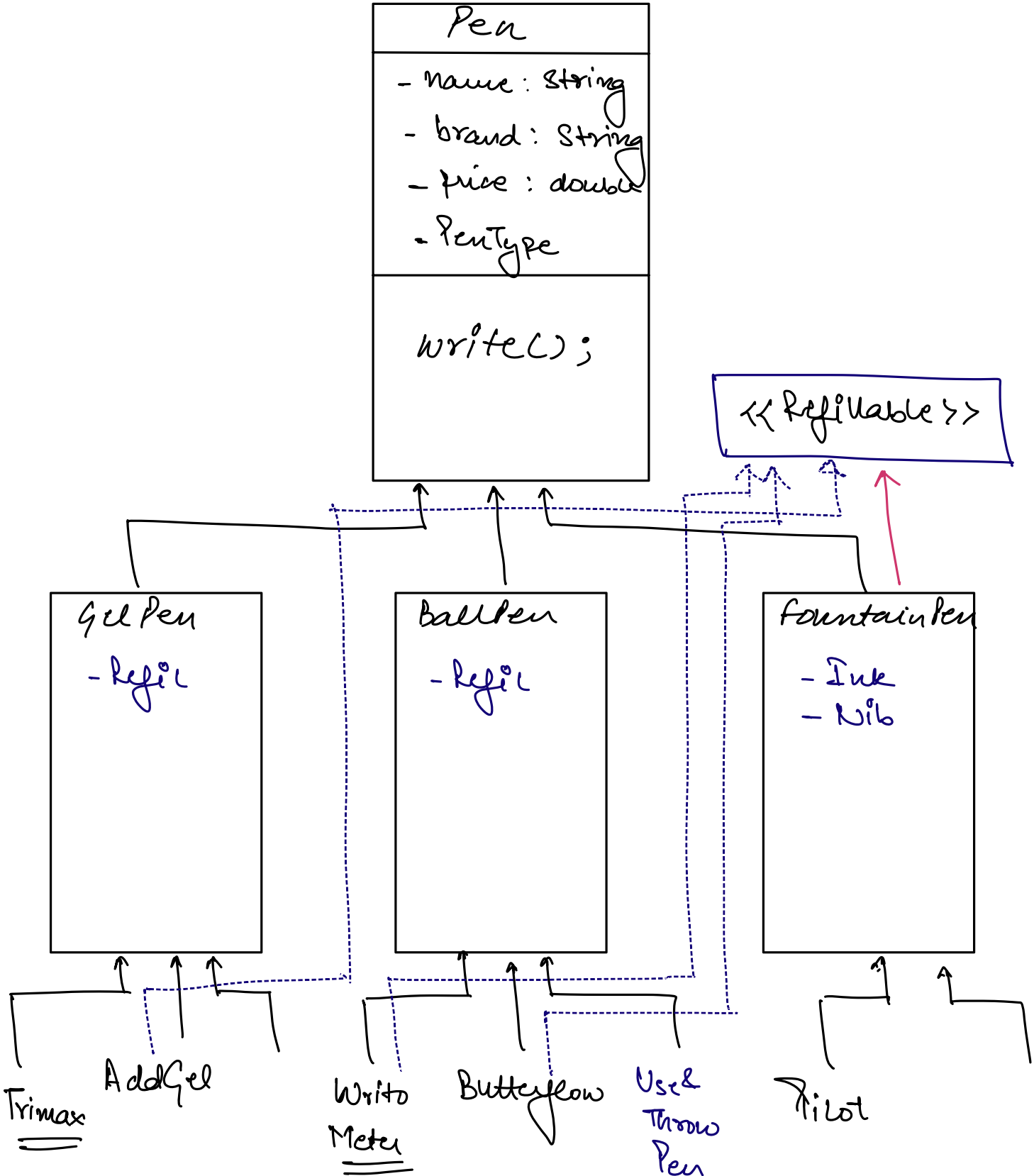
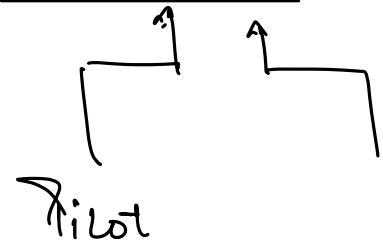
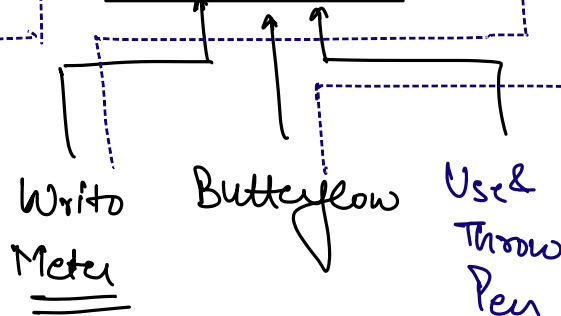
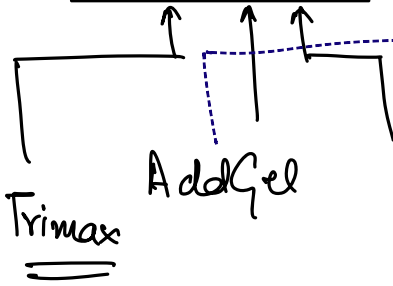
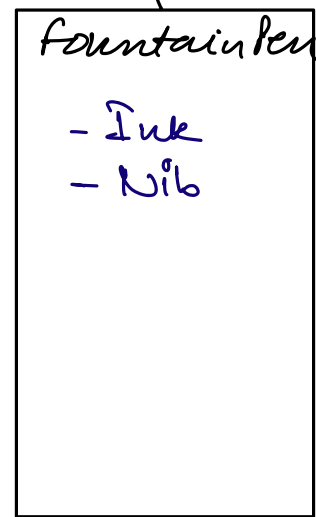
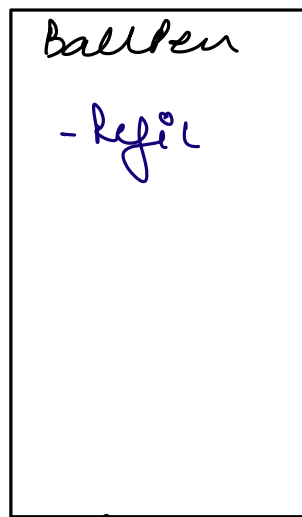
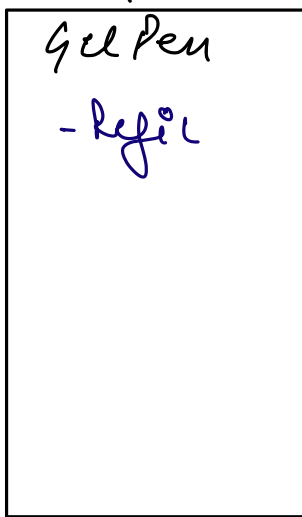
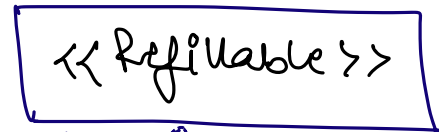
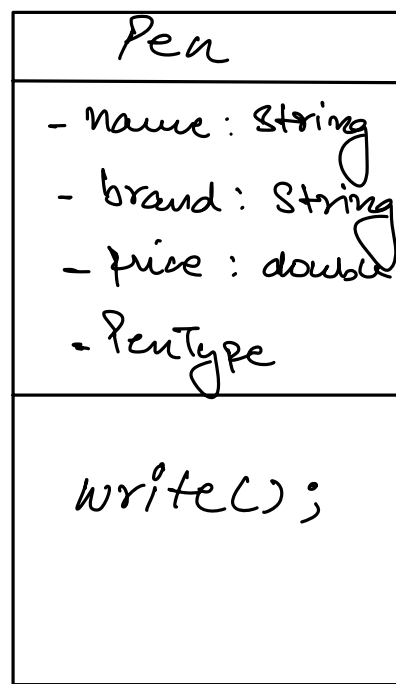
- 1) Any physical entity that can write is a Pen.
- 2) Different type of pens: ball pen, Gel pen & fountain pen.
- 3) Ball & Gel pen will have a refil, fountain pen won't have a refil. Refil will have Ink.
- 4) Fountain pen will directly have Ink.
- 6) Ink can be of different colors.
- 7) Refil will have a Nib.
- 8) Fountain pen will have Nib directly.
- 9) Nib can be of different radius.
- 10) Some pens can be refilled & some pens can't.



Class Diagram

① Visualize the system.  Physical Structure.
User Journey

② find out all the nouns in the requirements,
Create a class if we need to store data
for this noun in the DB.



PenType
BALL, GEL, <u>FOUNTAIN.</u>

```

Pen p = new BallPen()
        GelPen()
        FountainPen()

```

```

if (p.getPenType().equals("—"))

```

```

=> Pen p = Penfactory.getPen(⊙)

```

Refil
Ink Nib

Ink
Color List<feature>

Nib
radius

feature
TRANSPARENT, FRAGRANCE, ERASABLE, WATER-PROOF, — — —

— * —